

Shooting Bullets

Now let's add the ability to shoot bullets. We'll write code that fires a bullet, which is represented by a small rectangle, when the player presses the spacebar. Bullets will then travel straight up the screen until they disappear off the top of the screen.

Adding the Bullet Settings

At the end of the `__init__()` method, we'll update *settings.py* to include the values we'll need for a new `Bullet` class:

```
settings.py    def __init__(self):
               --snip--
               # Bullet settings
               self.bullet_speed = 2.0
               self.bullet_width = 3
               self.bullet_height = 15
               self.bullet_color = (60, 60, 60)
```

These settings create dark gray bullets with a width of 3 pixels and a height of 15 pixels. The bullets will travel slightly faster than the ship.

Creating the Bullet Class

Now create a *bullet.py* file to store our `Bullet` class. Here's the first part of *bullet.py*:

```
bullet.py    import pygame
             from pygame.sprite import Sprite

             class Bullet(Sprite):
                 """A class to manage bullets fired from the ship."""

                 def __init__(self, ai_game):
                     """Create a bullet object at the ship's current position."""
                     super().__init__()
                     self.screen = ai_game.screen
                     self.settings = ai_game.settings
                     self.color = self.settings.bullet_color

                     # Create a bullet rect at (0, 0) and then set correct position.
                     ❶ self.rect = pygame.Rect(0, 0, self.settings.bullet_width,
                                             self.settings.bullet_height)
                     ❷ self.rect.midtop = ai_game.ship.rect.midtop

                     # Store the bullet's position as a float.
                     ❸ self.y = float(self.rect.y)
```

The `Bullet` class inherits from `Sprite`, which we import from the `pygame.sprite` module. When you use sprites, you can group related elements in your game and act on all the grouped elements at once. To create a bullet instance, `__init__()` needs the current instance of `AlienInvasion`, and we call